

Mohammed Aman Jukaku

Engineering Student

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Profile

Information Science Engineering student with hands-on experience building end-to-end software systems across **AI/ML, data engineering, backend development, and cloud infrastructure**. Experienced in developing and deploying **machine learning applications, data-processing pipelines, automation workflows, and backend services**.

Education

Information Science Engineering, R V College of Engineering 09/2024 – Present
Bangalore, Karnataka
• CGPA: 8.4/10 | Year: 3rd | Semester: 5th

Internships

Technical Staff Intern, Soket AI 07/2026 – Present

- Building **LLM-agent-driven pipelines for programmatic 3D scene generation**, combining 3D frameworks and AI agents to automatically construct structured scenes.
- Developed a **game-tools ecosystem** for interactive puzzle domains, integrating game logic, tool execution, and synthetic interaction-data generation for LLM training.

Data Curation Intern, Indian Institute of Technology Gandhinagar 03/2026 – 06/2026

- Built automated **LLM data pipelines operating at hundreds-of-millions-of-samples scale**, spanning data collection, scraping, cleaning, normalization, deduplication, filtering, annotation, and validation.
- Developed synthetic-data workflows for **single-turn and multi-turn instruction-tuning datasets**, including schema standardization and automated validation.
- Analyzed conversational datasets for **response quality, duplicate samples, token/sequence distributions, domain balance, and structural consistency** for downstream LLM training.

Software Developer Intern, Emertxe Information Technologies 05/2024 – 09/2024
Bangalore, Karnataka

- Fine-tuned **OpenAI foundation models** for **domain-specific tasks** and integrated model inference into application workflows.
- Built a **full-stack platform** for **testing, managing, and deploying fine-tuned LLMs**.
- Developed an AI mock-interview system that automatically **generates interviews, conducts sessions, analyzes responses, and produces scored evaluation reports**.

Projects

Transformer Language Model From Scratch, PyTorch · Transformer Architecture · BPE 06/2025

- Implemented a **7.7M-parameter autoregressive Transformer from scratch**, including multi-head self-attention, custom BPE tokenization, and the end-to-end training pipeline.
- Trained on a text corpus using a **32K-token vocabulary, 256-token sequence length, and 10,000 training steps**.
- Built an interactive web demo to **inspect generated outputs and experiment with learned contextual relationships**.
- **GitHub:** <https://github.com/Amjuks/LLM-Experimental> 

Decentralized Multimodal Biometric Identity System, Deep Learning · IPFS · Ethereum · AES-256 01/2026

- Built a multimodal identity-verification system combining **facial recognition, voice verification, and identity-document analysis**.
- Encrypted biometric embeddings using **AES-256-CBC**, stored them through IPFS, and used **Ethereum Sepolia smart contracts** for identity mappings and verification records.
- Evaluated the system on **50 users**, achieving **97.2% facial recognition, 89.6% voice verification, 92.4% document verification, and 90.8% combined multimodal accuracy**, with a 2% false-acceptance rate.
- Reduced approximately **4 MB of raw biometric data to ~12 KB for IPFS storage** and a 32-byte blockchain representation.
- **GitHub:** <https://github.com/Amjuks/Identity-Verification-System-Blockchain> 

Skin Lesion Classification, PyTorch · CNN · Grad-CAM · HAM10000 12/2025

- Developed a **3.2M-parameter CNN** for seven-class skin-lesion classification on HAM10000, using class-weighted training to address dataset imbalance.
- Achieved approximately **85% validation accuracy** and integrated **Grad-CAM** to visualize model attention for predictions.
- **Kaggle:** <https://www.kaggle.com/code/amjuks/skin-cancer-10k-v2/> 

AI Smart Surveillance Robot, ESP32 · YOLO · Computer Vision · Embedded Systems

- Built an autonomous surveillance robot using **ESP32, Wi-Fi, ultrasonic sensing, DC motors, and a custom 3D-printed chassis**.
- Integrated **YOLO object detection and target tracking** with autonomous navigation and tracking-loss alerts.

- Developed a real-time web dashboard with **live video, target selection, robot controls, event logs, alerts, and navigation mapping.**
- **Github:** <https://github.com/Amjuks/AI-Smart-Surveillance-Robot-> 

BumpYourServer, *Python · discord.py · CI/CD · Production Systems*

- Built and operate a production Discord bot serving communities representing approximately **100,000 members.**
- Implemented server promotion, moderation/reporting, configuration, and a **cross-server global chat system.**
- Managed asynchronous event handling, persistent data, deployment, CI/CD, reliability, and production operations.

 Skills

Languages	AI/ML	Backend & Data	Infrastructure
Python, C, JavaScript	PyTorch, TensorFlow, Hugging Face, Transformers, CNNs, GANs, Computer Vision, LLMs, OpenCV, scikit-learn	Django, Flask, REST APIs, PostgreSQL, Firebase, NumPy, Pandas	Git/GitHub, Linux, Docker, Kubernetes, CI/CD, GCP

 Achievements & Certifications

- **Winner**, College-wide Researchathon, Skin Lesion Classification
- **1st Place**, Maze-Solving Robotics Competition, BMS College of Engineering
- **Certified Paragliding**
- **Harvard CS50x, CS50AI, CS50W**
- **Google Cloud Skills Boost**
- **chess.com Puzzle Rank: Top 10%**